

Exploring the Prognostic Potential of Let-7 MicroRNA in Clear Cell Renal Cancer

Sanchit Raizada¹, Ivan Raizada², Linfeng Cai³

¹ Government of Alberta, Calgary, Alberta, Canada

²Webber Academy, Calgary, Alberta, Canada

³ Hong Kong Baptist University, Hong Kong SAR

Student Authors:

Ivan Raizada: Grade 8 Webber Academy

15 SUMMARY

16 Kidney cancer remains one of the most widespread strains of cancer, with clear cell renal cell
17 carcinoma (ccRCC) being among the most prevalent, aggressive, and deadly forms. Despite
18 this, early, non-invasive detection remains a challenge. MicroRNAs (miRNAs), including the
19 let-7 family, are small non-coding RNAs (sncRNAs). It is established that they alter gene
20 expression and frequently show abnormal regulation in cancer. However, existing studies
21 disagree on whether let-7 expression increases or decreases in ccRCC. To address this gap,
22 we examined whether let-7 is positively or negatively correlated with ccRCC and identified
23 which let-7 microRNA might serve as the most effective biomarker. We hypothesized that let-
24 7 would generally decrease in ccRCC, based on its behaviour in other cancers. We also
25 hypothesized that let-7a or let-7b would be the most promising biomarker based on their
26 detectability in urine supernatant and other cancers. Using patient data from the Genomic
27 Data Commons (GDC), we applied statistical analysis to measure direction and significance
28 of regulation. Unexpectedly, we documented an increase in let-7 expression in ccRCC,
29 contradicting the prevailing understanding and our hypothesis. This suggests that the role of
30 let-7 in ccRCC may differ from its role in other cancers. Despite this unexpected result, let-7b
31 did demonstrate the most significant expression difference between normal and cancerous
32 tissue, especially in the early stages. A machine learning model trained on let-7b expression
33 achieved high classification accuracy, showing promise for early, non-invasive ccRCC
34 detection. These findings contribute to resolving existing conflicts in the literature and highlight
35 let-7b's potential as a diagnostic biomarker.

37 INTRODUCTION

38 Worldwide, approximately 263,500 instances of ccRCC are diagnosed annually, making it the
39 most prevalent subtype of renal cell carcinoma (RCC) (1). RCC accounts for more than 90%
40 of all kidney cancer cases, and ccRCC constitutes 75–80% of RCC cases (1). Early-stage
41 ccRCC is asymptomatic and can remain undetected until it has already progressed to an
42 advanced stage (1). It is a primary tumour that begins in the kidneys and does not originate
43 from the spread of other organs. It can be hereditary, but it usually stems from unknown
44 causes. Diagnosis often begins with imaging techniques such as computed tomography (CT)
45 scans or ultrasounds, followed by a biopsy or MRI (2).

46 The conventional diagnostic approaches are limited in several ways. Firstly, CT scans and
47 ultrasounds may not always be able to distinguish kidney masses from cancer. They are also
48 limited in their sensitivity to small tumours. Ultrasounds are also operator-dependent, thus

making them less reliable. Biopsies have a chance of yielding a non-diagnostic result; 10-20% of kidney biopsies do not yield a definitive diagnosis, resulting in a longer delay and potentially requiring additional biopsies. All these limitations can lead to a lowered prognosis for patients with ccRCC (3-5).

Early and accurate cancer detection is critical for improving patient outcomes, underscoring the need for reliable biomarkers (6). This is where microRNAs (miRNAs) emerge as potential solutions. MiRNAs are short non-coding RNAs that regulate gene expression by inhibiting protein production from target messenger RNAs (mRNAs), an intermediary RNA that carries genetic information from DNA to the ribosome for protein synthesis. The disabling of mRNA molecules is done through the RNA-Induced Silencing Complex (RISC). RISC components, such as Argonaute proteins, bind miRNAs to target specific mRNAs through sequence complementarity, primarily between the 3' Untranslated Region (3'UTR) of the mRNA and the miRNA seed region with imperfect base pairing (8). MiRNAs can silence multiple mRNAs by preventing ribosome attachment or promoting mRNA degradation. This inactivation of mRNAs results in a post-transcriptional silencing to effectively control or fully deactivate specific proteins. The miRNAs produced by the cell, and thus the proteins silenced, differ significantly between healthy and diseased tissues (7).

Among the miRNAs, some act as tumour suppressors, silencing genes that promote cell proliferation and inhibit a cancer's growth and maturation (9). They are usually underexpressed in higher stages of cancers. Others function as oncogenes by silencing genes that inhibit proliferation and drive the progression of cancer (9). Thus, they are usually overexpressed in higher stages of cancers (9). For example, miR-210 is shown to be upregulated in later phases of ccRCC, while miRNAs such as miR-141 are strongly downregulated in later phases of ccRCC (9). Therefore, the aforementioned dysregulation of miRNAs has been consistently shown to contribute to cancer development (9).

One of the most researched miRNAs is the let-7 family of miRNAs (10). These were also among the first miRNAs to be discovered in *Caenorhabditis elegans* (10). This widely known family is also well established as a tumour suppressor because let-7 is known to inhibit many oncogenes that support cell proliferation and cancer stem cell (CSC) behaviour. For example, let-7 is known to inhibit well-known oncogenes such as Kirsten rat sarcoma viral oncogene homolog (KRAS), cellular myelocytomatosis oncogene (c-MYC), High Mobility Group AT-hook 2 (HMGA-2), and others (10).

Specifically in kidney physiology, let-7 is known for regulating normal renal function by controlling the progression of the cell cycle and tissue homeostasis (11). For example, they are known to suppress known oncogenes and fibrotic pathways (those which drive scar tissue formation), such as the TGF- β signalling pathway (12, 13). Let-7 is also known to be involved in nephrogenesis, with higher levels signalling the termination of nephrogenesis (14). Although many let-7 members share common seed regions and overlapping biological functions, individual members often also have their own niches as well and may have abnormalities in regard to their regulation in cancer (13). In particular, let-7b's regulation has been shown to be context-dependent (10, 13). This distinctive behaviour suggests let-7b has a unique regulatory role in cancer.

In ccRCC, let-7 is generally known to be downregulated, supporting its widely accepted tumour-suppressive role (13). Gowrishankar et al. looked at a total of 99 samples and found that most let-7 members were downregulated compared to adjacent normal tissue, with let-7b approaching statistically significant downregulation (9). Interestingly, they observed that a couple of let-7 members were upregulated; let-7g, let-7i, and let-7f were upregulated compared to the normal tissue (9). In general, the study documented the status quo, suggesting that let-7 is a tumour suppressor.

However, there was a study by Fedorko et al. that showed contrasting results (11). In that study, urine supernatant from kidneys with ccRCC was analyzed, and all six let-7 family members examined were found to have increased expression. While not directly comparable and not disproving one another per se, it nonetheless raises questions on let-7's behaviour within the kidney tissues of ccRCC patients. To further clarify this uncertainty, it warrants more study to use tissue-level data to classify the regulation of let-7 members.

To address this research gap, we analyzed expression data from kidney tissue by using the Cancer Genome Atlas (TCGA) project from the Genomic Data Commons Database (GDC). We hypothesized that let-7 expression would decrease in the samples from ccRCC patients, as most studies typically accept this (10). We also hypothesized that let-7a and let-7b would exhibit the largest differences in expression, as similar results were reported in a study analyzing urine supernatant in ccRCC as well as other studies in other cancers (11, 15, 16).

To test our hypotheses, we extracted the expression from all let-7 members from the TCGA-KIRC project in the GDC and then graphed the data as well as doing an independent two-sampled t-test to see the trend and the significance of the regulation. We found that let-7 expression overall increased, with only let-7c and let-7e having decreased. Notably, let-7b showed the largest difference in expression between normal and cancerous tissue, supporting its potential as a biomarker. This finding implies that in a clinical setting, the best let-7 miRNA to detect ccRCC may be let-7b. Early detection using miRNA biomarkers like

let-7b could improve prognosis for ccRCC patients by offering a less invasive and more accurate diagnostic tool.

RESULTS

Let-7 has a well-established association with ccRCC, and we explored its use for earlier diagnosis of ccRCC (13). We aimed to determine if let-7 would be downregulated or upregulated overall. To evaluate this, we quantified the expression of let-7 with the unit “reads per million miRNA mapped” (RPM). This adjusts the raw count (the number of times a particular miRNA is detected) for the total reads in the sample (it is at least 1 million in the TCGA-KIRC, the database used in the present study; this is necessary for the accuracy and quality of the miRNA quantification data).

The miRNA concentration data was obtained from the TCGA-KIRC dataset and graphed the data of the let-7 members across the stages of cancer. There were 20 normal samples and a total of 537 cancer samples spread across the 4 stages of cancer. We looked at all the let-7 members available to us and presented their expression levels. We determined the significance of the differences between normal and cancer samples by obtaining the p-value for each let-7 member with a two-sample independent t-test.

For a more fine-grained understanding of let-7 miRNAs’ stage-dependent variations, we made box charts of average expression level versus cancer stage. These diagrams help identify stage-specific variations in let-7 expression and pinpoint potential thresholds where expression levels significantly change. From the box plot of let-7b, we could see that let-7b expression minimally varied between the cancer stages but increased dramatically from the normal to cancer stages (**Figure 1**). To summarize the results and the trends in the data, a violin plot was constructed, representing all 11 let-7 miRNAs tested and their varying expression.

On the application of the t-test, the majority of let-7 family members, like let-7a, showed a slight increase in cancer samples with a p-value approaching statistical significance (Figure 2). Let-7f, -7g, and -7i saw a statistically meaningful increase. Let-7c showed a decrease approaching statistical significance. However, this result was likely due to the presence of three large outliers, which can be seen in the let-7c scatterplot, and was evaluated using the z-score (**Appendix**). Let-7d saw relatively no change, and let-7e saw a statistically significant decrease. Out of all let-7 miRNAs tested, let-7b had the largest difference between the normal and cancerous samples, with the lowest p-value (1.18×10^{-7}). We found that across all stages of ccRCC, let-7b's expression was significantly different compared to the normal tissues.

Comparing adjacent groups, there was a highly significant difference between normal tissues and Stage 1 of ccRCC, a difference approaching statistical significance between Stage 1 and Stage 2, a moderately significant difference between Stage 2 and Stage 3, and no statistically significant difference between Stage 3 and Stage 4 (**Table 1**). The above results indicated that let-7b would be the best biomarker out of the let-7 family.

To demonstrate this, we constructed a Support Vector Machine (SVM) model in Python to predict the tissue status (cancerous or normal) based on the expression level of let-7b. An SVM model is a supervised predictive machine learning model used in research to differentiate between multiple classes for predictive purposes. We created an SVM model using the scikit-learn (or sklearn) library, a popular library for machine learning in Python. The SVM model was implemented using the Support Vector Classifier (SVC) in scikit-learn. In order to mitigate the present class imbalance in the data, which could lead to misleading metrics and poor generalizability, we applied oversampling. Oversampling, or the artificial and intelligent addition of samples to the minority class, was employed using the popular technique known as Synthetic Minority Oversampling Technique (SMOTE).

Upon training and oversampling, using scikit-learn's built-in evaluation methods, we evaluated and tabulated the results and metrics of our model (**Table 2**). Relevant metrics for evaluation include accuracy, precision, recall, and Receiver Operating Characteristic-Area Under Curve (ROC-AUC). Accuracy, or the ratio of correct over total predictions, indicates how often the SVM model assigns the correct class; however, it is subject to overfitting biases. Precision is the accuracy over only the positive class (ccRCC). Recall (or sensitivity) is the accuracy over only members of the negative class (healthy). ROC-AUC gives the probability that the SVM model ranks a random positive sample higher than a random negative sample (**Table 2**). ROC-AUC's primary advantages are that it accounts for class imbalance and it quantifies separation ability.

In the classification report for the SVM, evaluated on a dataset of 215 samples, the model predicted 195 samples correctly and only 20 samples incorrectly. The SVM had a precision of 0.93 and a recall of 0.88 for "Normal" samples. Furthermore, for "Cancer" samples, the model had a precision of 0.89 and a higher recall of 0.94 (**Table 4**). The overall accuracy of the model was 0.91, indicating that 91% of all predictions were correct (**Table 4**). The harmonic mean of the precision and recall is defined as the F-1 score. This statistic gives a balanced look at the two ways of measurement. Both macro and weighted averages for F1-score, recall,

and precision are equal to 0.91, suggesting balanced performance across both classes. The model shows slightly better recall for "Cancer" and better precision for "Normal" (**Table 4**).

DISCUSSION

Let-7 is known for its potential for diagnostic purposes (10). We aimed to determine whether let-7 would be upregulated or downregulated in ccRCC. The analysis showed that, between normal and cancerous tissues, the expression of let-7 members mainly increased (e.g. let-7b), while some decreased (e.g., let-7c), and others were not correlated with ccRCC (e.g., let-7d). The let-7 member with the largest difference between normal and cancerous tissues was let-7b.

For a model to be effective at diagnosing a tissue as cancerous, the separation between normal and cancer classes should be large. Additionally, the concentration should be adequately high (in the thousands of RPM) for the miRNA to be easier to detect and distinguish in sequencing data. This is because the higher the expression level, the more frequently the sequence occurs in the sequencing data, increasing the signal-to-noise ratio (18). A signal-to-noise ratio measures how strong the actual signal (desired sequence) is compared to the background noise. The high signal-to-noise ratio stemming from the high concentration of the desired sequence makes it easier to detect from the background noise. This facilitates confident statistical analysis and robust identification, which is critical for both research and clinical applications.

The TCGA-KIRC database used in the present study was used in earlier studies. They showed that high expression of let-7 members correlated with lower survival for ccRCC, and vice versa (10, 19). This was evaluated using a tool called Prognostic MicroRNA Tool (PROGmiR) (10, 19). PROGmiR is designed to help researchers investigate the prognostic significance of miRNAs across various cancer types using publicly available data sets. The study used the survey of all 33 human cancer databases from TCGA and Gene Expression Omnibus (GEO) that had been uploaded to PROGmiR and showed that across all cancers, the upregulation of the majority of let-7 members was associated with better prognosis, supporting its generally accepted tumour suppressor role (10, 19). In ccRCC specifically, all let-7 members were positively correlated with prognosis. The only exceptions to this statement were let-7g and let-7f, which showed no statistically significant correlation, and let-7i was negatively correlated with prognosis. These observations suggested that let-7 is a tumour suppressor in most cancers, including ccRCC. PROGmiR does not train a prediction algorithm from scratch but rather analyzes existing survival data using standard statistical methods like Cox proportional

221 hazards models and Kaplan-Meier survival curves (19). This allows researchers to form
222 hypotheses about the tumour-suppressive or oncogenic roles miRNAs may play in cancer (19).

223 While both their study and ours used the TCGA-KIRC dataset, the structure of the data used,
224 the questions asked, and the analysis methods were different. PROGmiR only analyzed
225 samples from cancer patients to determine whether higher or lower levels of let-7 were
226 associated with better prognosis (19). In contrast, we analyzed expression data from both
227 'Normal' and 'Cancer' tissue types to identify whether let-7 members were upregulated or
228 downregulated in ccRCC (10, 17). This direct comparison between healthy and tumour tissue
229 allowed us to assess differential expression more clearly. In addition, our methodology was
230 relatively simple—two clearly defined sample types (normal vs. cancerous) were directly
231 compared to determine expression changes—whereas PROGmiR's predictive survival
232 analysis was built on statistical models within the cancer-only group (19).

233 This distinction is important. Our study focused on resolving whether let-7 is upregulated or
234 downregulated in ccRCC and determining which let-7 member has the greatest biomarker
235 potential for detecting ccRCC. Their study focused on patient survival, using only cancer
236 samples and Kaplan-Meier-based statistical predictions, with the goal of assessing whether
237 relative differences in miRNA expression levels among cancer patients correlated with clinical
238 outcomes (19). Their scope was the whole tumour and patient survival; ours was the tumour
239 versus normal comparison to identify expression patterns (10, 19).

240 Therefore, while our findings might seem to conflict with PROGmiR's, they are answering
241 different questions. For example, we found that let-7b might be upregulated in cancer tissue
242 compared to normal tissue, and the earlier study showed that, among cancer patients, those
243 with even higher expression of let-7b might have better survival (19). These two outcomes are
244 not contradictory and could be complementary. Our goal was to classify tissue as cancerous
245 or not based on the expression of let-7 members, while theirs was to predict patient survival
246 based on relative miRNA levels within the cancer group (19). These differences in sample
247 subsets, analytical approach, scope, and interpretation explain why both studies can use the
248 same dataset and still arrive at different—but not mutually exclusive—conclusions. One
249 potential explanation could be a compensatory tumour-suppressor response. The development
250 of cancer creates oncogenic stress, which may result in an increased counter-regulatory
251 expression of tumour suppressors by the cell to counteract the development of cancer (20). This
252 would explain why tumours have upregulated tumour suppressors that may be involved in such
253 pathways. However, with the artificial addition of tumour suppressors, it is able to mitigate and
254 perhaps reverse the effects of cancer, consistent with PROG-miR's predictions.

Our initial hypothesis was that a decrease in let-7 members would occur overall, as let-7 is a tumour suppressor and this result was observed in many other papers (10, 17-19, 21, 22). However, this hypothesis was not borne out. There was a 2.63-fold increase in expression of let-7b between the normal and cancer stages on average. This suggests that in ccRCC, let-7b might act as an oncogene—an uncommon idea. We found let-7b to have the lowest p-value and most significant increase; however, let-7f, 7g, and -7i showed highly significant increases as well. This result is consistent with the findings of Fedorko et al. (11). However, in our study, let-7a showed a decrease approaching statistical significance, while let-7e showed a highly statistically significant decrease, with the other miRNAs showing statistically insignificant regulations. These results diverge from what was observed in the urine, where Fedorko observed an increase in all let-7 members. This highlights the need for further investigation for an adequate explanation.

The upregulation of let-7b observed is unusual, as let-7 members are commonly classified as tumour suppressors. This behaviour warrants further comparison. A review by Boyerinas et al. (2010) documented that let-7b was upregulated in gastrointestinal stromal tumours (GIST) and in lymphoma (23). In lymphoma specifically, Lawrie et al. (2008) found let-7b elevated in DLBCL and follicular lymphoma cases of the lymph nodes that subsequently underwent high-grade transformation to diffuse large B-cell lymphoma (24). This finding is consistent with the possibility that let-7b promotes cancer progression. Elevated let-7b expression has also been detected in the urine supernatant of ccRCC patients as previously discussed, matching our tissue-level finds (11). The authors proposed that tumour cells may passively shed this miRNA more, though the mechanism was not specified. In blood cancers, Yazarlou et al. (2021) reported let-7b upregulation in AML bone marrow compared to normal controls (25). Collectively, let-7b upregulation has been demonstrated in kidney tissue, B-cell lymph nodes, soft tissue, and bone marrow, challenging the current tumour-suppressive notion and suggesting a more context-dependent role of let-7 members. The mechanisms causing this variability remain to be definitively described, and a more systematic survey across cancer types would be beneficial to determine whether the upregulation observed in ccRCC is truly cancer-dependent or a shared phenomenon.

We have also shown let-7b as a promising candidate for a model identifying a tissue as cancerous or not. Since the p-value was far below the threshold, a model based on the let-7b expression would perform very well. The model we constructed, an SVM model, had a high accuracy, F1 score, and ROC-AUC score, with all benchmarks approximately 0.9, showing comparable and slightly better performance than a similar SVM model made by Li et al. (Tables 3 and 4) (27). Their model used 23 unnamed genes, unlike our sole usage of let-7b, to also

predict stages of ccRCC samples. They had more samples than our study (604 compared to our total of 557 samples) and achieved an accuracy of 81.1% and an AUC of 0.86, comparable to our 90.6% accuracy and 0.93 ROC-AUC score (**Table 2, 4**) (27). This suggests that let-7b is a viable biomarker for cancer with potential for more advanced neural networks in the future, and could allow the classification of tissues using let-7b to be a reliable and mainstream form of identification, enabling cancer patients to benefit from early identification and more dependable and less invasive results (**Tables 3, 4**).

Despite our efforts, there are some inherent limitations to this study. While necessary, the use of SMOTE to oversample normal samples may have introduced unintended biases, primarily to the minority class (here, the normal samples) (28, 29). A large class imbalance was also present in this study: 20 normal samples compared to 537 cancer samples. While this was accounted for during machine learning by implementing SMOTE, this imbalance must still be considered in the application of our results. Our study also only tested models trained on a single miRNA, rather than a multi-featured SVM model, and exclusively used SVM rather than other machine learning algorithms such as random forest or logistic regression. This limits the confidence with which we can quantify how effective let-7 miRNAs are as biomarkers, and future studies would be needed to identify which model best predicts diagnosis based on let-7 expression.

The cohort used was also limited to TCGA. While this resulted in uniformity and reduced the risk of confounding factors, it is a limitation that must be taken into account, as any unintended biases in TCGA would be reflected in our data and could potentially compromise our results. In addition, the SVM model was tested on samples from the same cohort on which it was trained. We therefore did not use an independent cohort to assess the generalizability of the model, and any metrics obtained are only directly applicable to this cohort.

Furthermore, since these machine learning models were trained on data sourced from kidney tissues, real-world implementation directly using ours or similar models would still require a biopsy to obtain the input sample, undermining the goal of a less-invasive approach. Our study has shown that let-7b is likely the best single biomarker for ccRCC in kidney tissues; however, for real-world implementation, future studies should retrain the model using urine supernatant to determine whether our findings remain consistent and which model performs best. Such a model would be better suited for non-invasive diagnosis. The diagnostic potential of let-7b is supported by both the observed expression data and the machine learning model's performance, suggesting that let-7 expression could in the future serve as a tool for determining whether a patient has ccRCC. However, let-7b may also be upregulated in other

RCC subtypes, which would limit the specificity of individual biomarkers for a definitive diagnosis, highlighting the need to further investigate the use of multiple biomarkers in combination. Further research should be done to determine how specific this behaviour is to let-7b compared to comparable RCC cancers and kidney cancers as a whole. The merit of this diagnosis only presents itself if let-7b's behaviour is highly dependent on cancer, allowing for models to be developed to differentiate between them. One potential implementation of diagnosis based on urine supernatant is outlined below.

First, urine samples are taken, and, using RNA extraction kits, RNA, including miRNA, is then isolated. The RNA can be cleaned, and using an miRNA-specific microarray chip that includes let-7b probes, the expression of let-7 can be quantified (11, 30). Using a machine learning model, the probability of ccRCC can be given. With this method, the presence of ccRCC can be identified earlier and non-invasively, potentially improving patient outcomes. While the proposed method is theoretically viable, further optimization and validation studies are likely to yield more effective diagnostic approaches. In the future, work could be done to put the method above into practice and check the results.

In conclusion, let-7b's average expression between normal and cancerous samples more than doubled, which is unexpected as let-7b usually acts as a tumour suppressor. However, this dramatic increase allows it to be a promising biomarker candidate, and to show this, the SVM model thus created had a 90.7% accuracy. In the future, better models, better methods for getting let-7 expression from patients, and more research on the unusual behaviour of let-7b's expression are needed. When these questions are answered, the role of let-7 will be better understood, and ccRCC can be better identified and prevented.

MATERIALS AND METHODS

Acquiring the Data

The Genomic Data Commons (GDC) was used to obtain the data. From the platform, the ccRCC data were downloaded. The miRNA levels of 557 patients were downloaded: 20 normal tissues and 537 cancerous, spread across Stage 1 to Stage 4. The data were filtered to include transcriptome profiling miRNA-Seq data, specifically miRNA Expression Quantification data. The data used were open access and were filtered to include samples whose primary site was the kidney and the diagnosis was ccRCC.

Once all the cancer files were downloaded, the filter was switched to "normal", thus including only healthy solid kidney tissue as controls. As such, 20 normal cases were downloaded, as

356 SMOTE would later be used to oversample the data due to limited variability between the
357 cases.

358 The organization of the folders is detailed in the appendix. This organization is necessary, as
359 it allowed for an efficient way to structure the spreadsheet and acquire data from the files,
360 such as a file's universally unique identifier (FileUUID) and the ID of the case, or CaseID.
361 When rgrep or grep -r obtained the instance from the file, it also returned the path it took. By
362 renaming the files in this way, when rgrep returned the path it took, all relevant information in
363 the file can be obtained, facilitating data compilation.

364 *Extracting the Data*

365 To extract the data, Terminal's grep functionality was used to extract the line with the target
366 miRNA. In the .txt files with the expression data, the data for each miRNA are all on one line.
367 Rgrep or grep -r would prove useful as it could display all the essential information at once.
368 Grep -r recursively checks for the target string in any text files. This means that the command
369 grep -r "has-let-7a"* can be run to get information from all the samples for let-7a, allowing for
370 easy retrieval.

371

372 For each let-7 member, the text output from the command was copied and pasted into the
373 Sublime Text editor. This allowed for formatting the text so that it could be pasted into the
374 spreadsheet. In the spreadsheet, each sheet corresponds to a let-7 family member. The
375 spreadsheet showed the stage of the sample (normal or cancer stage). It also showed the
376 CaseID and the ID of the person from whom the sample was taken. Inputting this into the GDC
377 returns all the files related to that person, including the miRNA expression from the kidneys.
378 The FileUUID, when entered into the GDC portal, provides access to the file from which the
379 data was taken. The spreadsheet also contains the name and URL of the file for easy access.

380 *Analysis of the Data*

381 For every miRNA sheet, there were two graphs. One was a dot plot of every sample: RPM vs.
382 Stage. The other was a bar graph of the mean RPM vs. Stage (**Appendix**). Using an
383 independent two-sample t-test, the p-value for each miRNA was obtained, and it was found
384 that let-7b had the smallest p-value. After that, conclusions were drawn, and the rest of the
385 results are presented in the Results section.

386 *Construction of the SVM model*

387 We used the Pycharm IDE by JetBrains to code the SVM model. The SVC class from the
388 scikit-learn library was used to develop an SVM classification model. Then, using the pandas
389 library, key data columns—including Stage and RPM—were extracted from the spreadsheet.
390 Normal samples were oversampled using SMOTE to address the imbalance in sample sizes.

391 Synthetic Minority Oversampling Technique (SMOTE) is a popular method for oversampling
392 datasets when limited data is available (28). It was selected because its assumptions aligned
393 well with the characteristics of the dataset. The data was relatively large (~500 samples), one-
394 dimensional (RPM only), and the classes were well separated for let-7b (shown by the low p-
395 value). All of these characteristics point to choosing SMOTE as the oversampling of choice
396 (28). Since SMOTE does not perform well with regression problems, it was suitable here
397 because the problem being addressed was classification (28). With no significant drawbacks,
398 SMOTE was chosen as the best approach for oversampling.

399 After applying SMOTE, the data were split into training and testing sets, and the model was
400 trained accordingly. The trained model was saved as a .joblib file to optimize efficiency,
401 enabling reuse without repeated computation. Predictions were made using .predict and
402 .predict_proba, which provided both class predictions and associated confidence probabilities.

403 *Possible Errors in Application of Methodology*

404 Possible errors could have arisen during data extraction with grep, leading to misidentified or
405 missed miRNA entries. Misnaming or disorganizing files, such as CaseID mismatches, could
406 have referenced incorrect samples. Additionally, manual data entry into spreadsheets is prone
407 to typographical errors, and mistakes in statistical calculations, such as p-value determination
408 or graphing, could misrepresent findings. As such, efforts were made to ensure that all
409 statistical results were legitimate.

410 Double-checking procedures were implemented to minimize these potential errors, and
411 outliers were validated by cross-referencing sample data with RPM values. File organization
412 and naming conventions were carefully verified to ensure accurate data matching. Human and
413 technical errors were reduced by inspecting statistical results for anomalies and checking data
414 transfer processes. Future improvements could involve directly importing .txt files into the
415 spreadsheet to eliminate manual transfer errors and leveraging additional datasets to enhance
416 model accuracy and robustness.

417 A better way to avoid these errors is to directly input the .txt files with the RPM values into the
418 spreadsheet, avoiding any transfer errors. Using more samples from websites other than the

419 GDC would further improve the accuracy of the model and establish more concrete trends
420 between normal and cancer expression levels for let-7b.

421 Another limitation of my study is that there is a possibility of unknown factors affecting the
422 expression of let-7b. As these factors are not controlled, they could result in bias or skewing
423 in the data. As such, the limited knowledge on let-7b is a limiting factor of this research.

424
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544 **Figures:**

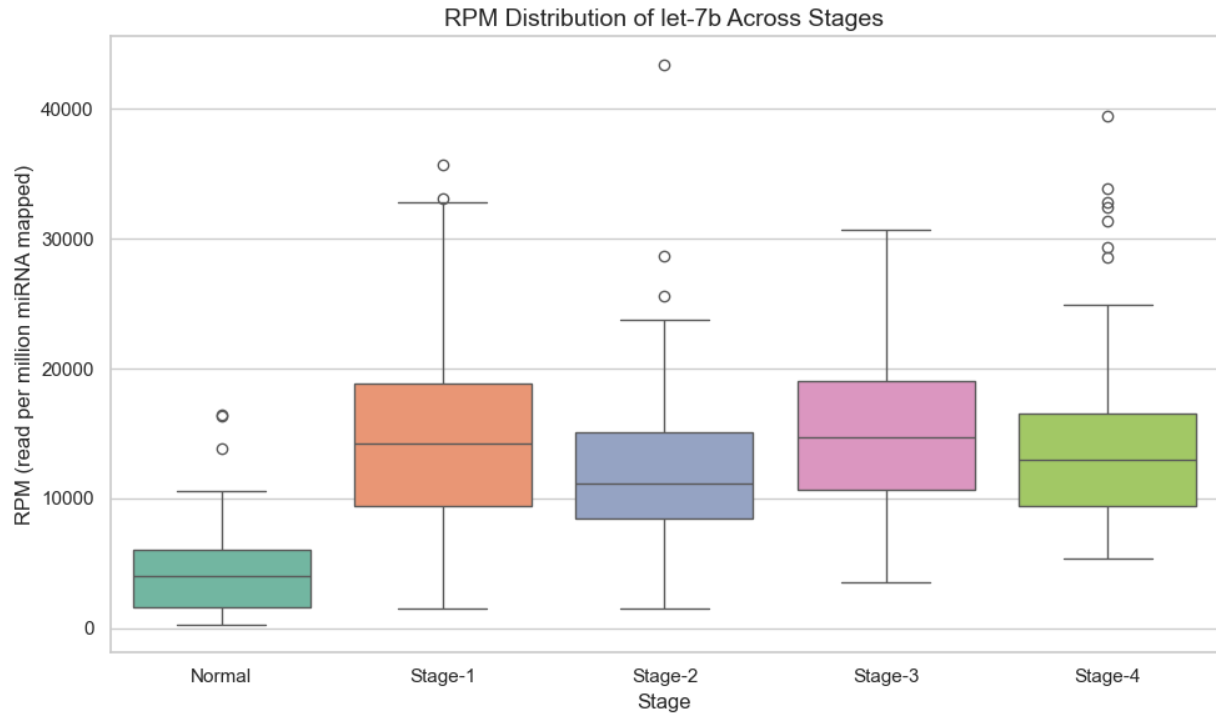


Figure 1: Box plot of mean RPM of let-7b at different cancer stages. The number of samples for each group: Normal samples: 20, Stage 1: 242, Stage 2: 65, Stage 3: 130, Stage 4: 100. It can be seen that the Normal samples have lower variance compared to cancer samples, whom have longer bars. Outliers have also been plotted, and it can be seen that most are positive outliers, meaning that they have larger expression than the mean, while there are no outliers below the bars.

Tables:

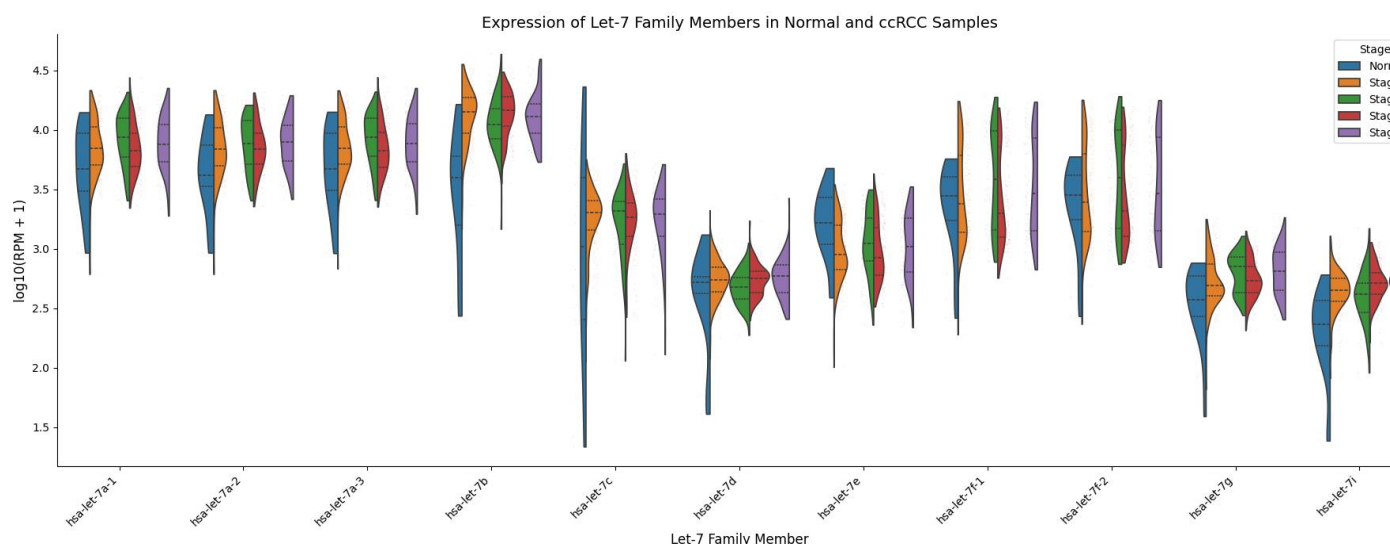


Figure 2. Expression levels of let-7 in Normal kidney tissue and across the four stages of ccRCC. In this violin plot, normalized expression levels [$\log_{10}(\text{RPM} + 1)$] were graphed for 11 let-7 family members in Normal tissue and across the stages of ccRCC samples from Stages I–IV. The width of each violin represents the density of samples at a given expression level. Several let-7 family members, including let-7a isoforms, let-7b, let-7f-1, let-7f-2, let-7g, and let-7i, exhibit higher expression in tumour samples compared with normal tissue. Let-7b demonstrates one of the most pronounced increases across tumour stages. Let-7d exhibited a minimal change in regulation, while let-7c and let-7e were lowered in ccRCC. Overall, expression patterns suggest differential regulation of let-7 family members during ccRCC progression.

	Stage 1	Stage 2	Stage 3	Stage 4
Normal	2.74 (<0.001)	2.40 (<0.001)	2.80 (<0.001)	2.61 (<0.001)
Stage 1		0.88 (0.055)	1.02 (0.675)	0.95 (0.387)
Stage 2			1.16 (0.034)	1.09 (0.291)
Stage 3				0.93 (0.251)

Table 1. P-values and fold changes for let-7b Expression Between Normal Tissue and

Cancer Stages 1–4. This table shows pair-wise p-values between normal tissue and each cancer stage studied (1 – 4) in brackets, with the fold change being represented outside the brackets. P-values <0.001 indicate very significant differences, while values closer to the alpha value of 0.05 indicate moderately significant to approaching meaningful statistical differences. Values over the threshold indicate no statistical significance. Values in the diagonal from top left to bottom right indicate statistical significance from one stage to the next, and can be used to assess how strongly let-7b varies with ccRCC linearly.

Metric	Value
Accuracy	0.907
Precision	0.887
Recall	0.936
ROC-AUC	0.931

Table 2. Overall Performance Metrics on SVM Model. This table shows a brief overview of key diagnostics of the SVM model. Accuracy measures the proportion of total predictions that the model got right over all predictions the model made. Precision determines out of all instances that were predicted as positive and were correctly classified. Recall determines out of all actual positive instances, what proportion the model was able to catch. Receiver Operating Characteristic – Area Under Curve, or ROC-AUC, quantifies the model’s overall ability to segregate the classes.

	Predicted Normal	Predicted Cancer
Actually Normal	93	13
Actually Cancer	7	102

Table 3. Confusion Matrix of the SVM model with correct and incorrect predictions. This table details the number of false positives (13), false negatives (7), true positives (102) and true negatives (93). False positives are when the SVM predicts that the result is positive, when it is in fact negative. False negatives are when the SVM predicts that the result is negative, but it is actually positive. True positives and true negatives are when the model

correctly predicts the outcome. The information in this table can then be used to determine statistics such as accuracy and precision of the SVM model.

Class	Precision	Recall	F1-Score	Support
Normal	0.93	0.88	0.9	106
Cancer	0.89	0.94	0.91	109
Accuracy	0.91	215		
Macro Avg	0.91	0.91	0.91	215
Weighted Avg	0.91	0.91	0.91	215

Table 4: Classification Report with specifics about the SVM model. This table gives the precision, recall, F1-Score, and support, separated by class. Precision measures the proportion of true positives to all positives predicted. Accuracy is the ratio of correct predictions to total predictions. F1 score is the harmonic mean of precision and recall, able to encapsulate both values in a clean metric. Support refers to the number of samples in each class.

APPENDIX

This is the hyperlink for the entire sheet of the let-7 members used:

https://docs.google.com/spreadsheets/d/1OxaipNosENnGiObiOAnu9GqviznvuBl8RHOFIvrl7Ro/edit?usp=drive_link .

File Organization:

FULL_DATA

↳ Stages (Ex, Normal, Stage 1, Stage 2, etc.)

↳ Unzipped the folder of the sample, renamed to CaseID

↳ A folder containing .txt, named after the FileUUID column

↳ .txt file with the expression of the miRNAs, the name is FileName

This is the code for the model: [https://github.com/IvanR2625/ccrcc_svm_model]

